

Madhura Kale

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EDUCATION

AISSMS Institute of Information Technology

B.Tech. in Computer Engineering | CGPA: 7.34/10

Pune, India

Aug 2023 – Jun 2027

MP International School and Junior College

Higher Secondary Education | 73.67%

Pune, India

Jun 2021 – Mar 2023

Kendriya Vidyalaya Ganeshkhind

Secondary Education | 90%

Pune, India

Jun 2020 – Mar 2021

SKILLS

Languages: Python, C++, SQL

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Git

Databases: SQL, PostgreSQL, MySQL, SQLite, SQLAlchemy

Web Development: Flask, FastAPI, Django Fundamentals, REST APIs, React.js, HTML, CSS, Streamlit, Postman

AI / Machine Learning: Machine Learning, Deep Learning, NLP, RAG, LangChain, Semantic Search, FAISS, TensorFlow, OpenCV, MediaPipe, Transfer Learning

Data & Analytics: Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Data Visualization, Power BI, Tableau, Excel

Tools: Git, GitHub, Linux Fundamentals, VS Code

PROJECTS

DocuMind AI – RAG-Based Multi-Document Question Answering System

Python, LangChain, FAISS, Google Generative AI, Groq, Streamlit

- Built an end-to-end Retrieval-Augmented Generation (RAG) application enabling natural-language question answering over multiple uploaded PDF documents.
- Implemented document ingestion, text chunking, and embedding generation with Google's text-embedding-004, plus semantic similarity search using FAISS.
- Integrated Llama 3 through Groq to generate context-grounded responses using retrieved document content, with prompt constraints to reduce hallucinations.

Student Performance Prediction

Python, Pandas, Scikit-learn, Machine Learning, Flask, REST APIs

- Designed and validated a regression pipeline covering data cleaning, feature engineering, and exploratory data analysis; benchmarked Linear Regression, Random Forest, and CatBoost models.
- Achieved a final R^2 score of 0.692 with an average prediction error of 5–6 marks, and deployed the model through a Flask REST API, validating endpoints with Postman.

IBM HR Employee Attrition Prediction

Python, Pandas, Scikit-learn, Power BI

- Analyzed and cleaned data from 1,470 employee records across 35 attributes, built and compared classification models, and addressed class imbalance to identify key attrition drivers.
- Developed an interactive Power BI dashboard to translate analytical findings into actionable business insights.

Sign Language Detection System

Python, OpenCV, MediaPipe, Scikit-learn

- Built a real-time computer vision system tracking 21 hand landmarks per frame for ML-based gesture classification, optimized for low-lag predictions from live webcam input.

PUBLICATION

A Deep Learning-Based Web Application for Plant Leaf Disease Classification Using MobileNetV2 | IJCRT

- Published a peer-reviewed research paper presenting a MobileNetV2-based transfer learning framework, and developed a web application that classifies plant leaf diseases from uploaded images.

ACHIEVEMENTS

- Problem Solving: Solved 200+ Data Structures and Algorithms problems on LeetCode across Easy, Medium, and Hard difficulty levels.
- Leadership: Coordinated 8+ technical and cultural events as Secretary of iConnect across two semesters, managing logistics and team collaboration for events with 100+ attendees.
- Hackathon Achievement: Cleared Round 1 of the Advitya Hackathon at IIT Bombay by presenting a business idea and advancing to the next stage of the competition.